

# An Analysis of Receiver operating characteristic

## Receiver operating characteristic

### Receiver operating characteristic -- Part 1

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**History** The ROC curve was first used during World War II for the analysis of radar signals before it was employed in signal detection theory. Following the attack on Pearl Harbor in 1941, the United States military began new research to increase the prediction of correctly detected Japanese aircraft from their radar signals. For these purposes they measured the ability of a radar receiver operator to make these important distinctions, which was called the Receiver Operating Characteristic. In the 1950s, ROC curves were employed in psychophysics to assess human (and occasionally non-human animal) detection of weak signals. In medicine, ROC analysis has been extensively used in the evaluation of diagnostic tests. ROC curves are also used extensively in epidemiology and medical research and are frequently mentioned in conjunction with evidence-based medicine. In radiology, ROC analysis is a common technique to evaluate new radiology techniques. In the social sciences, ROC analysis is often called the ROC Accuracy Ratio, a common technique for judging the accuracy of default probability models. ROC curves are widely used in laboratory medicine to assess the diagnostic accuracy of a test, to choose the optimal cut-off of a test and to compare diagnostic accuracy of several tests. ROC curves also proved useful for the evaluation of machine learning techniques. The first application of ROC in machine learning was by Spackman who demonstrated the value of ROC curves in comparing and evaluating different classification algorithms. ROC curves are also used in verification of forecasts in meteorology.

### Receiver operating characteristic -- Part 2

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**History** The ROC curve was first developed by electrical engineers and radar engineers during World War II for detecting enemy objects in battlefields, starting in 1941, which led to its name ("receiver operating characteristic"). It was soon introduced to psychology to account for the perceptual detection of stimuli. ROC analysis has been used in medicine, radiology, biometrics, forecasting of natural hazards, meteorology, model performance assessment, and other areas for many decades and is increasingly used in machine learning and data mining research.

### Receiver operating characteristic -- Part 3

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**Z-score** If a standard score is applied to the ROC curve, the curve will be transformed into a straight line. This z-score is based on a normal distribution with a mean of zero and a standard deviation of one. In memory strength theory, one must assume that the zROC is not only linear, but has a slope of 1.0. The normal distributions of targets (studied objects that the subjects need to recall) and lures (non studied objects that the subjects attempt to

recall) is the factor causing the zROC to be linear. The linearity of the zROC curve depends on the standard deviations of the target and lure strength distributions. If the standard deviations are equal, the slope will be 1.0. If the standard deviation of the target strength distribution is larger than the standard deviation of the lure strength distribution, then the slope will be smaller than 1.0. In most studies, it has been found that the zROC curve slopes constantly fall below 1, usually between 0.5 and 0.9. Many experiments yielded a zROC slope of 0.8. A slope of 0.8 implies that the variability of the target strength distribution is 25% larger than the variability of the lure strength distribution. Another variable used is  $d'$  ( $d'$  prime) (discussed above in "Other measures"), which can easily be expressed in terms of z-values. Although  $d'$  is a commonly used parameter, it must be recognized that it is only relevant when strictly adhering to the very strong assumptions of strength theory made above. The z-score of an ROC curve is always linear, as assumed, except in special situations. The Yonelinas familiarity-recollection model is a two-dimensional account of recognition memory. Instead of the subject simply answering yes or no to a specific input, the subject gives the input a feeling of familiarity, which operates like the original ROC curve. What changes, though, is a parameter for Recollection ( $R$ ). Recollection is assumed to be all-or-none, and it trumps familiarity. If there were no recollection component, zROC would have a predicted slope of 1. However, when adding the recollection component, the zROC curve will be concave up, with a decreased slope. This difference in shape and slope result from an added element of variability due to some items being recollected. Patients with anterograde amnesia are unable to recollect, so their Yonelinas zROC curve would have a slope close to 1.0.

### Receiver operating characteristic -- Part 4

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Plots of the four results above in the ROC space are given in the figure. The result of method A clearly shows the best predictive power among A, B, and C. The result of B lies on the random guess line (the diagonal line), and it can be seen in the table that the accuracy of B is 50%. However, when C is mirrored across the center point (0.5,0.5), the resulting method  $C'$  is even better than A. This mirrored method simply reverses the predictions of whatever method or test produced the C contingency table. Although the original C method has negative predictive power, simply reversing its decisions leads to a new predictive method  $C'$  which has positive predictive power. When the C method predicts p or n, the  $C'$  method would predict n or p, respectively. In this manner, the  $C'$  test would perform the best. The closer a result from a contingency table is to the upper left corner, the better it predicts, but the distance from the random guess line in either direction is the best indicator of how much predictive

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power a method has. If the result is below the line (i.e. the method is worse than a random guess), all of the method's predictions must be reversed in order to utilize its power, thereby moving the result above the random guess line.

### **Receiver operating characteristic -- Part 5**

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The contingency table can derive several evaluation "metrics" (see infobox). To draw a ROC curve, only the true positive rate (TPR) and false positive rate (FPR) are needed (as functions of some classifier parameter). The TPR defines how many correct positive results occur among all positive samples available during the test. FPR, on the other hand, defines how many incorrect positive results occur among all negative samples available during the test. A ROC space is defined by FPR and TPR as x and y axes, respectively, which depicts relative trade-offs between true positive (benefits) and false positive (costs). Since TPR is equivalent to sensitivity and FPR is equal to  $1 - \text{specificity}$ , the ROC graph is sometimes called the sensitivity vs  $(1 - \text{specificity})$  plot. Each prediction result or instance of a confusion matrix represents one point in the ROC space. The best possible prediction method would yield a point in the upper left corner or coordinate (0,1) of the ROC space, representing 100% sensitivity (no false negatives) and 100% specificity (no false positives). The (0,1) point is also called a perfect classification. A random guess would give a point along a diagonal line (the so-called line of no-discrimination) from the bottom left to the top right corners (regardless of the positive and negative base rates). An intuitive example of random guessing is a decision by flipping coins. As the size of the sample increases, a random classifier's ROC point tends towards the diagonal line. In the case of a balanced coin, it will tend to the point (0.5, 0.5). The diagonal divides the ROC space. Points above the diagonal represent good classification results (better than random); points below the line represent bad results (worse than random). Note that the output of a consistently bad predictor could simply be inverted to obtain a good predictor. Consider four prediction results from 100 positive and 100 negative instances: